-ANTI-GASTRITIS AND ANTI-GASTRIC ULCER DRUG –

MUCOSTA® Tablets 100

< Rebamipide>

Designated Drug and Prescription-only Drug

Storage: Store below 30^{o}C.
Expiration date: Three years after the date of manufacture. (The expiration date is indicated on the package.)

CONTRAINDICATIONS
(MUCOSTA Tablets are contraindicated in the following patients)
Patients with a history of hypersensitivity to any ingredient of this drug.

DESCRIPTION

1. Composition
MUCOSTA Tablets 100:
Each MUCOSTA Tablet contains 100 mg of rebamipide and the following inactive ingredients: microcrystalline cellulose, low substituted hydroxypropylcellulose, hvdroxypropylcellulose, magnesium stearate, hypromellose, macrogol 6000, and titanium oxide.

2. Product Description
MUCOSTA Tablets 100 are white film-coated tablets.

Appearance	Diameter (mm)	Thickness (mm)	Weight (mg)	Code
	8.1	3.4	Approx. 175	OG33

INDICATION

- Gastric ulcers
- Treatment of gastric mucosal lesions (erosion, bleeding, redness and edema) in the following conditions: acute gastritis and acute exacerbation of chronic gastritis.

DOSAGE AND ADMINISTRATION

• Gastric ulcers: The usual adult dosage of rebamipide is 100mg (1MUCOSTA Tablet) taken by the oral route three times daily, in the morning, in the evening, and before at bedtime.
•Treatment of gastric mucosal lesions (erosion, bleeding, redness, and edema) in the following conditions; acute gastritis and acute exacerbation of chronic gastritis: The usual adult dosage of rebamipide is 100 mg (1 MUCOSTA Tablet) three times daily taken by the oral route.

PRECAUTIONS

1. Adverse Reactions
Of 10,047 patients treated, adverse reactions, including abnormal laboratory findings, were reported in 54 patients (0.54%). Of 3,035 patients aged over 65 years, adverse reactions were noted in 18 patients (0.59%). The nature and incidence of adverse reactions were not different between the same in elderly and younger patients. The following summary of data includes adverse reactions voluntari1y reported after marketing (Figures are total cases reported at the time of approval and at the completion of reexamination of MUCOSTA Tablet 100)

(1)Clinically significant adverse reactions

1) Shock, anaphylactoid reactions (incidence unknown*): Shock or anaphylactoid reactions may occur. Patients should therefore be closely monitored. If abnormal findings are observed, the drug should be discontinued and appropriate measures taken.

2) Leukopenia (incidence <0.1%) and thrombocytopenia (incidence unknown*): Leukopenia and thrombocytopenia may occur. Patients should therefore be closely monitored. If abnormal findings are observed, the drug should be discontinued and appropriate measures taken.

3) Hepatic dysfunction (incidence <0.1 %) and jaundice (incidence unknown*): Hepatic dysfunction and jaundice, as indicated by increases in AST (GOT), ALT (GPT), γ-GTP, and alkaline phosphatase levels, have been reported in patients receiving MUCOSTA Tablets. If abnormal laboratory findings are observed, the drug should be discontinued and appropriate measures taken

(2) Other adverse reactions

Body system/frequency	<0.1%	*Incidence unknown
Hypersensitivity (note 1)	Rash, pruritus, drug-eruption-like eczema, other symptoms of hypersensitivity	Urticaria
Neuro-psychiatric		Numbness, dizziness, Sleepiness
Gastro-Intestinal	Constipation, feeling of abdomen enlarged, diarrhea, nausea, vomiting, heartburn, abdominal pain, belching, taste abnormality, etc.	Thirst
Hepatic(note 2)	Increased AST (GOT), ALT (GPT), γ-GTP, alkaline phosphatase levels	
Hematologic	Leukopenia, granulocytopenia, etc.	Thrombocytopenia
Other	Menstrual disorders, increased BUN levels, edema, feeling of a foreign body in the pharynx	Breast swelling and pain, gynecomastia, induction of lactation, palpitations, fever, facial flushing, numbness of tongue, cough, respiratory distress, alopecia

Note 1) If such symptoms of hypersensitivity occur, the drug should be discontinued.
Note 2) If transaminase levels are markedly increased or fever and rash develop, the drug should be discontinued and appropriate measures should be taken.
*The incidence rates of voluntarily reported adverse reactions are not known.

2. Drug Interactions
Rebamipide had little inhibitory effect on cytochrome P450 enzymes *in vitro*. Rebamipide is hardly metabolized in the liver and excreted in urine as an unchanged compound; thus, the drug has not much influence on metabolization of other drugs. Regarding the concomitant administration with other drugs with higher protein binding rats (H_2 blocker and Warfarin), although Mucosta (rebamipide) has a high protein binding rate (95% or more), its blood concentration is as low as up to 200 ng/mL; therefore, this drug is considered less likely to increase the b1ood concentration in competition with other drugs. In addition, there have been reports that Mucosta (rebamipide) has not been influenced by (1) plasma salicylic acid concentration (human) at the time of aspirin administration, (2) serum indomethacin concentration (human) at the time of indomethacin administration, and (3) plasma and tumor tissue 5-FU concentration (rat) at the time of UFT administration. No drug interactions between rebamipide and other drugs are reported.

3. Use in the Elderly
Special care is required in elderly patients to minimize the risk of gastrointestinal disorders, because these patients may be physiologically more sensitive to this drug than younger patients.

4. Use during Pregnancy, Delivery, or Lactation
(1) This drug should be administered to pregnant or possibly pregnant women only if the anticipated therapeutic benefit is thought to outweigh any potential risk. (The safely of this drug in pregnant women has not been established.)
(2) Nursing should be interrupted when this drug is administered to.
(Rat studies have shown that rebamipide is excreted in the breast milk).

5. Pediatric use
The safety of this drug in low birth weight infants, newborns, suckling infants, infants and children has not been established. (Clinical experience is insufficient).

6. Precautions for Use
Patient's Instructions for Use
Patients should be instructed not to ingest any portion of the press-through package (PTP). (There have been reports that the sharp edges of the sheet can cut or penetrate the esophageal mucosa if accidentally ingested, resulting in mediastinitis or other serious complications).

7. Overdose
In an efficacy study of gastric ulcer, one case of abdominal pain was reported among 81 patients who received rebamipide 900 mg/day (maximum dosage of the study). In post marketing experience, there is limited information on overdose. Treatment of overdose should be symptomatic. Close medical supervision and monitoring should continue until the patient recovering.

PHARMACOKINETICS
1. Plasma Concentrations
The table below shows the pharmacokinetic parameters of rebamipide following single oral administration of MUCOSTA Tablets 100 at the dose of 100 mg to 27 healthy male subjects in a fasted state.

Pharmacokinetic Parameters of Rebamipide

	t_{max} (hr)	C_{max} (ng/mL)	$t_{1/2}$ (hr)	AUC_{24h} (ng · hr/mL)
MUCOSTA Tablets 100	2.4±1.2	216±79	1.9±0.7	874±209

Mean value ±SD, n=27, $t_{1/2}$ calculated from values up to 12 hr.

The absorption of rebamipide following single oral administration at a dose of 150 mg to 6 healthy male subjects in a fed state tended to be slower than that in a fasted state. However, food did not affect bioavailability of the drug in humans. Pharmacokinetic parameters obtained from patients with renal impairment after single oral administration of rebamipide at 100 mg revealed higher plasma concentrations and a longer elimination half-life compared with those in healthy subjects. At steady state, rebamipide plasma concentrations observed in dialyzed renal patients following repeated administration were very close to the values simulated from single administration. Therefore, the drug was not considered to accumulate.

2. Metabolism
Rebamipide was primarily excreted as the unchanged compound in the urine after single oral administration to healthy adult males at a dose of 600 mg. A metabolite with a hydroxyl group at the 8th position was identified in the urine. However, the excretion of this metabolite was only 0.03% of the administered dose. The enzyme involved in the formation of the metabolite was CYP3A4.
(Note) The usual dosage in adults is 100 mg three times daily.

3. Excretion
Approximately 10% of the administered dose was excreted in the urine when rebamipide was administered as a single oral dose to healthy adult males at 100 mg.

4. Protein Binding
Rebamipide at 0.05 -5 μg/mL was added to human plasma *in vitro*, and 98.4%-98.6% of the drug was bound to plasma proteins.

CLINICAL STUDIES

1. Clinical Efficacy in Gastric Ulcer
MUCOSTA Tablets were studied in patients with gastric ulcer, using endoscopy for objective drug evaluation. In the final endoscopic assessment, the drug achieved complete healing in 60% (200/335) of the patients studied and near-complete healing in 67% (224/335). The clinical usefulness of this drug, based on efficacy and safety was demonstrated in a double-blind study. Six-month follow-up of 67 patients who showed healing at a daily dose of 300 mg revealed that recurrence occurred in only 4 patients (approx. 6%).

2. Clinical Efficacy in Acute Gastritis and Acute Exacerbation of Chronic Gastritis
MUCOSTA Tablets were studied in patients with acute gastritis or acute exacerbation of chronic gastritis. The drug achieved an 80% (370/461) global efficacy rate in the patients evaluated, with 76% (351 /461) showing moderate or marked improvement. The drug's clinical usefulness was found to be reproducible in a double-blind study.

PHARMACOLOGY
1. Experiments Using Animal Models

(1) Preventive or healing effects in gastric ulcer models
Rebamipide inhibited gastric mucosal injury in various experimental rat models of ulcers, including ulcers induced by water immersion restraint stress, aspirin, indomethacin, histamine, serotonin, and pyloric ligation. The drug also protected the mucosa from injury caused by other ulcerogenic conditions that presumably yield reactive oxygen species, including mucosal ischemia-reperfusion, administration of platelet activating factor (PAF) or diethyldithiocarbamate (DDC), and administration of indomethacin under stressed conditions. In a rat acetic acid-induced ulcer model, the drug promoted healing of gastric ulcers and was seen to suppress the recurrence and relapse of ulcers 120-140 days after ulcer induction.

(2) Preventive or healing effects in gastritis models
Rebamipide inhibited the development of taurocholic acid-induced gastritis and promoted healing of the mucosal inflammation associated with gastritis in rat experiments.

(3) Prostaglandin-increasing effect
Rebamipide increased the generation of prostaglandin E_2 (PGE_2) in the gastric mucosa in rats. The drug also increased the contents of PGE_2, 15-keto-13, 14-dihydro-PGE_2 (a metabolite of PGE_2) and PGI_2 in the gastric juice. In healthy male subjects, the drug again revealed the increasing effect on the PGE_2 content in the gastric mucosa and protected the gastric mucosa from injury caused by ethanol loading.

(4) Cytoprotective effect
Rebamipide exhibited a gastric cytoprotective effect to inhibit the mucosal injury induced by ethanol, strong acid, or strong base in rats. In *in vitro* studies, the drug also protected cultured gastric epithelial cells obtained from rabbit fetuses against aspirin- or taurocholic acid-induced injury.
In healthy male subjects, the drug inhibited gastric mucosal injury induced by aspirin, ethanol, or HCl-ethanol loading.

(5) Mucus-increasing effect
Rebamipide promoted gastric enzyme activity to synthesize high molecular weight glycoproteins, thickened the superficial mucous layer of gastric mucosa, and increased the amount of gastric soluble mucus in rats. Endogenous PGs were not involved in the increase in soluble mucus.

(6) Mucosal blood flow-increasing effect
Rebamipide increased gastric mucosal blood flow and improved impaired hemodynamics after blood loss in rats.

(7) Effect on mucosal barrier
Rebamipide did not ordinarily affect the gastric transmucosal potential difference in rats, but did inhibit lowering of the potential difference by ethanol.

(8) Effect on gastric alkaline secretion
Rebamipide promoted gastric alkaline secretion in rats.

(9) Effect on mucosal cell turnover
Rebamipide activated gastric mucosal cell proliferation and increased the number of covering epithelial cells in rats.

(10) Effect on gastric mucosal repair
Rebamipide restored the bile acid- or hydrogen peroxide-induced retardation of artificial wound-repair in cultured rabbit gastric epithelial cells.

(11) Effect on gastric secretion
Rebamipide did not alter either basal secretion of gastric juice or secretagogue-stimulated acid secretion.

(12) Effects on reactive oxygen species
Rebamipide scavenged hydroxyl radicals directly and suppressed superoxide production by polymorphonuclear leukocytes. The drug inhibited the gastric mucosal cell injury caused by reactive oxygen species released from neutrophils stimulated by *Helicobacter pylori in vitro*. The drug reduced the content of lipid peroxide in the gastric mucosa of rats treated with indomethacin under stressed conditions and inhibited the mucosal injury.

(13) Effect on inflammatory cell infiltration in the gastric mucosa
Rebamipide prevented inflammatory cell infiltration in rat models of taurocholic acid-induced gastritis and NSAID-induced or ischemia reperfusion-induced gastric mucosal damage.

(14) Effect on inflammatory cytokine release (interleukin-8) in the gastric mucosa
Rebamipide, taken by the oral route, suppressed the increased production of interleukin-8 in the mucosa of patients with *Helicobacter pylori*. The drug also inhibited the activation of NF-κB the expression of interleukin-8 mRNA, and the production of interleukin-8 in epithelial cells cocultured with *Helicobacter pylori* (*in vitro*).

PHYSICOCHEMISTRY

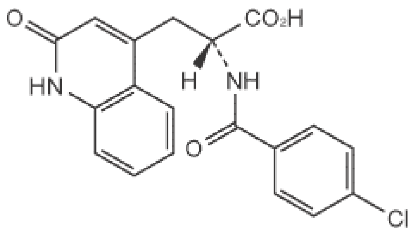
Non-proprietary name:
Rebemipide (INN)

Chemical name:
(2*RS*)-2-(4-chlorobenzoylamino)-3-(2-oxo-1,2-dihydroquinolin-4-yl) propanoic acid

Molecular formula:
$C_{19}H_{15}C_{l}N_{2}O_{4}$

Molecular weight:
370.79

Structural formula:



and enantiomer

Description:
Rebamipide occurs as a while crystalline powder. It has a bitter taste. It is very slightly soluble in methanol and ethanol (95%) and practically insoluble in water. Its *N*,*N*-dimethylformamide solution (1 in 20) shows no optical rotation.
Melting point: About 291 ^{o}C (decomposition)

PACKAGING
MUCOSTA Tablets 100:
Boxes of 100 tablets in press-through packages (PTP)

REQUESTS FOR LITERATURE SHOULD BE ADDRESSED TO:
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